

## No. of Functional Features Included in the Solution

|               |                                                 |
|---------------|-------------------------------------------------|
| Date          | 2 october 2026                                  |
| Team ID       | SWTID-2026-7456                                 |
| Project Name  | ComicCraft-AI Comic Creator using Gemini Models |
| Maximum Marks | 5 Marks                                         |

### Functional Features Overview:

This document lists the functional features included in the **ComicCraft - AI Comic Story Creator using Gemini Models** project. Each feature describes its purpose, implementation status and contribution to the overall solution.

| S.No | Feature Name              | Feature Description                                                                       | Module / Component            | Status (Done / In Progress / Pending) | Marks Contribution |
|------|---------------------------|-------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|--------------------|
| 1    | User Input                | Allows the user to enter story prompt, character name, setting, story tone and art style. | index.html / FastAPI          | Done                                  | High               |
| 2    | AI Story Generation       | Generates a structured comic story from the user's inputs using Gemini AI.                | gemini_flash.py / routes.py   | Done                                  | High               |
| 3    | Five-Panel Comic Creation | Converts the generated story into five individual comic panels.                           | routes.py / layout_builder.py | Done                                  | High               |
| 4    | AI Image Generation       | Generates illustrations for the comic panels using Hugging Face AI image generation.      | image_generator.py            | Done                                  | High               |
| 5    | Comic Preview             | Displays the generated comic panels with story content in the browser.                    | comic_preview.html / CSS      | Done                                  | High               |
| 6    | PDF Export                | Creates a downloadable PDF containing the generated comic.                                | exporters.py / routes.py      | Done                                  | Medium             |
| 7    | API Documentation         | Provides interactive API documentation for testing ComicCraft endpoints.                  | FastAPI /docs                 | Done                                  | Medium             |

|   |                |                                                                             |                                |      |        |
|---|----------------|-----------------------------------------------------------------------------|--------------------------------|------|--------|
| 8 | Error Handling | Handles API and image-generation errors and provides appropriate responses. | routes.py / image_generator.py | Done | Medium |
|---|----------------|-----------------------------------------------------------------------------|--------------------------------|------|--------|

Feature Category Breakdown:

| S. No | Category                  | Features in Category | Example Features                                     |
|-------|---------------------------|----------------------|------------------------------------------------------|
| 1     | User Interface (UI)       | 2                    | User Input, Comic Preview                            |
| 2     | Backend / Logic           | 3                    | Story Generation, Five-Panel Creation, Error         |
| 3     | Database / Storage        | 1                    | Generated comic images and PDF files stored locally  |
| 4     | API / Integration         | 2                    | Gemini AI Integration, Hugging Face Image Generation |
| 5     | Security / Authentication | 3                    | API keys stored securely using .env configuration    |

Feature Summary

| Metric                             | COUNT / VALUE |
|------------------------------------|---------------|
| TOTAL FEATURES PLANNED             | 8             |
| TOTAL FEATURES IMPLEMENTED         | 8             |
| CORE / MUST-HAVE FEATURES          | 5             |
| ADDITIONAL / NICE-TO-HAVE FEATURES | 3             |
| FEATURES TESTED & VERIFIED         | 8             |